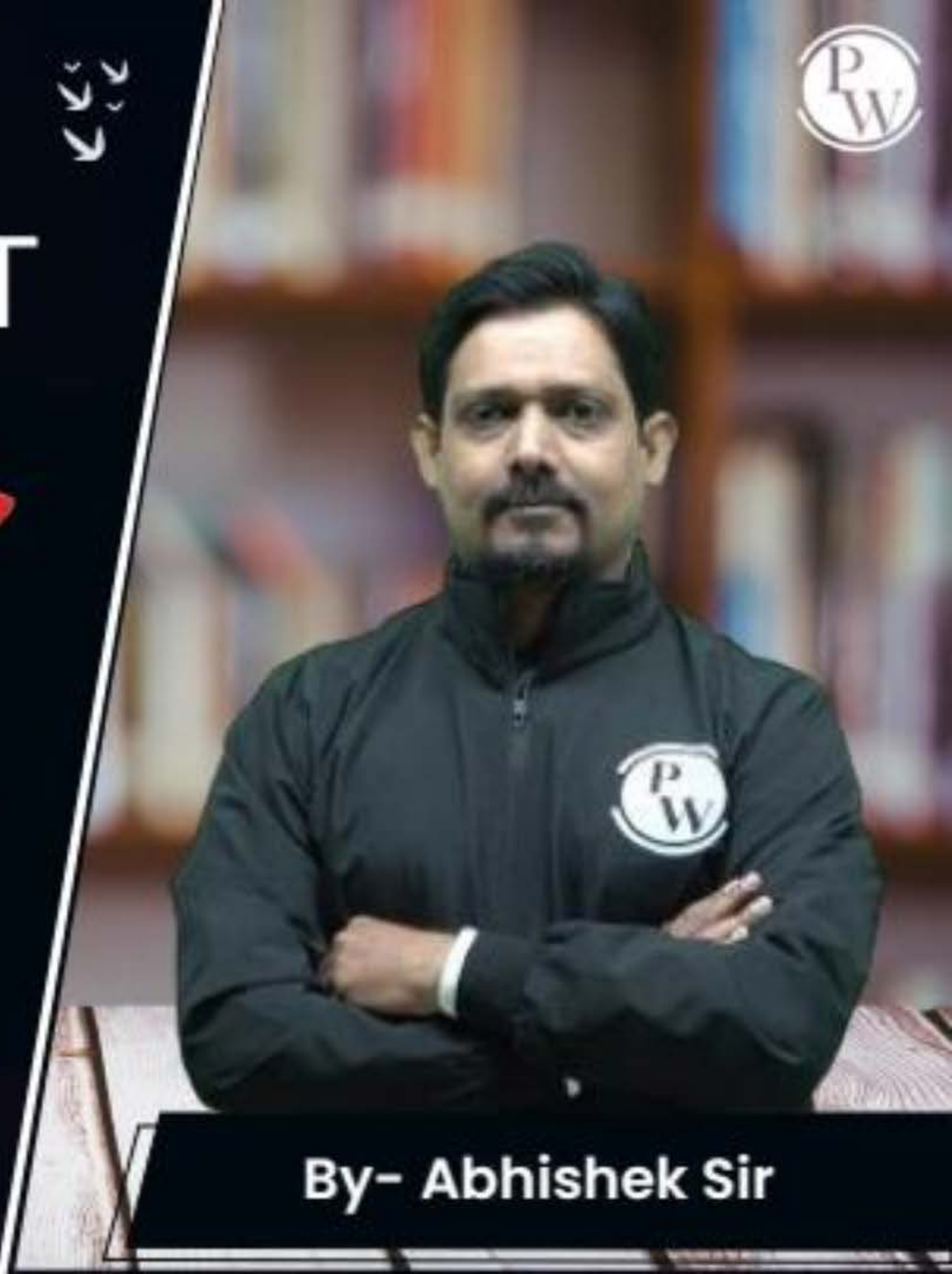
Computer Science & IT



Fundamental of C Language

Data Types & Operators

Lecture No. 05



By- Abhishek Sir

Recap of Previous Lecture



topic

topic

topic

topic

topic

char data type

long short

float data type

Slide

topics to be Covered



- topic — Arithmetic operation
- topic — Relational operator
- topic — Logical operator
- topic — Bit wise operator
- topic — ?: Condihonal operator

Slide



Topic : Operators & Arithmetic Expresion

Arithmetic expression :

1. integer & integer then result will be integer
2. int & float is float
3. float & float will be float

example 1. $\frac{13}{3} = 4$

example 2. $\frac{10.0}{4} = 2.5$

example 3 $\frac{10.0}{4.0} = 2.5$

Slide



Topic : Arithmetic Expression Rule

+ , - , * , / , %o , =

Binay operator applied on two operand.

Binay operator

e.g 5+10 operator

operand 5, 10

e.g. 5/3



Topic : Operators

% modulo operator

% - It gives remainder

$\frac{7}{3}$

7 - Dividend Quotient = 2

3 - Divisor Remainder = 1

Dividend = Quotient * divisor + Remainder

Slide



Topic : Operators

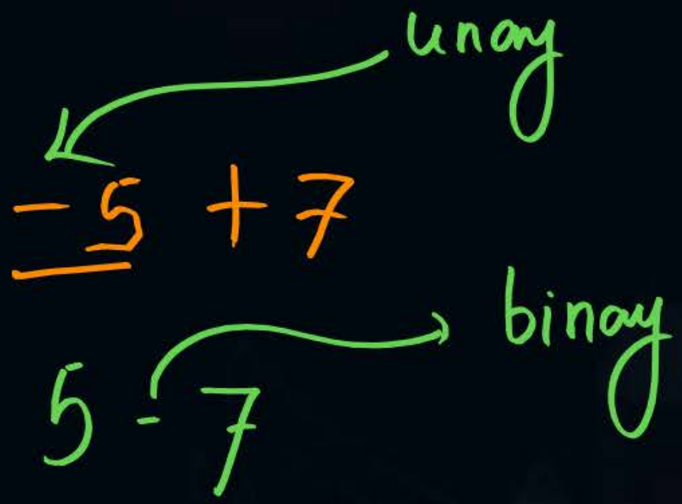
```
printf("%d", 7/3);
printf("%d", 7%3);
printf("%d", 7*3);
```

Slide



Topic : Operators

Unary operator : operator which is applied on Single operand



Slide



Topic : Operators

$5+6*7$

Rule

$= 5+42 = 47$

BODMAS

- B — Bracket
- O — order of
- D — Division
- M — Multiplication
- A — Addition
- S — Subtraction

precedence

order of evaluation of operators

Associativity

if precedence is Same then order of evaluation decided by associativity

$5+6+7$

Slide



Topic : Arithmetic Operator

+, -, *, /, %

Slide

Toipc: ++ and --

++
--

applied on variables

variable value is modified

does not applied on constant & expression

```
int x=5;
x++;
```

post increement

x=6

y=x++;

```
int x=5
++x;
```

pre increement

x=6

y=++x;

```
int y=10;
y--;
```

post decrement

y=9

x=y--

```
int y=10
--y;
```

pre decrement

y=9

x=--y

Slide



Toipc: Assigment Operator

```
int a=10;
```

a value assigned with 10

Slide



Toipc: Relational Operator

>	<	>=	<=	==	!=
Less than	Greater than	Greater than equal to	Less than equal to	Exactly equal	Not equal to

10>5 10<5 10>=5 10<=10 5==5 5!=6

Slide



Toipc: Relational Operator Example

```
#include<stdio.h>
int x = 40;
int  main() {
        printf("%d\n", 30>40);
        printf("%d\n", 30>=40);
        printf("%d\n", 30==40);
        printf("%d\n", 30!=40);
        printf("%d\n", 40!=30);
        printf("%d\n", 40==40);
        printf("%d\n", 50>50);
        printf("%d\n", 50<=50);
        return 0 ;
}
```

0
0
1
1
1
1
0
1

40 != 40 — 0

40 == 40 — 1

Slide



Toipc: Logical Operator

AND

A	B	A && B
0	0	0
0	1	0
1	0	0
1	1	1

$5<10 \;\&\&\; 3>1$

$1 \;\&\&\; 1$

OR

A	B	A || B
0	0	0
0	1	1
1	0	1
1	1	1

$5>10 \;||\; 1<6$

$0 \;||\; 1 = 1$

NOT (unary)

!

A	!A
0	1
1	0

Slide



Toipc: Logical Operator &&

Any Nonzero value is true

Zero is false

Slide



Toipc: Logical Operator (AND) &&

```
#include <stdio.h>
int main(void) {
    int a = 20;
    int b = 30;
    printf("%d", a > 10 && b > 10);
}
```

20 > 10 && 30 > 10

1 && 1 = 1

Slide



Toipc: Logical Operator (OR) ||

```
#include <stdio.h>
int main(void) {
   int a = 20;
   int b = 5;
   printf("%d", a > 10 || b > 10);

}
```

20>10 || 5>10

1 || 0 = 1

Slide



Toipc: Logical Operator (NOT) !

printf("%d", !a);

```
int a=5;
int b=10
printf("%d", !(a>b))
```

= !(5>10)

= !0

= 1

Slide



Bit-wise Operator

AND — &

OR — |

XOR — ^

Negation — ~

<< leftshift

>> Rightshift

Slide



Number System

Decimal Number ⟶ Binary No

XOR

a	b	a XOR b
0	0	0
0	1	1
1	0	1
1	1	0

Slide



Bit-wise Operator

Operators	Meaning of operators
&	Bitwise AND
|	Bitwise OR
^	Bitwise exclusive OR
~	Bitwise complement
<<	Shift left
>>	Shift right

Bit-wise Operator

a = 5

b = 4

a & b

00000101

00000100

00000100

a | b

00000101

00000100

00000101

~a

~(00000101)

= 11111010

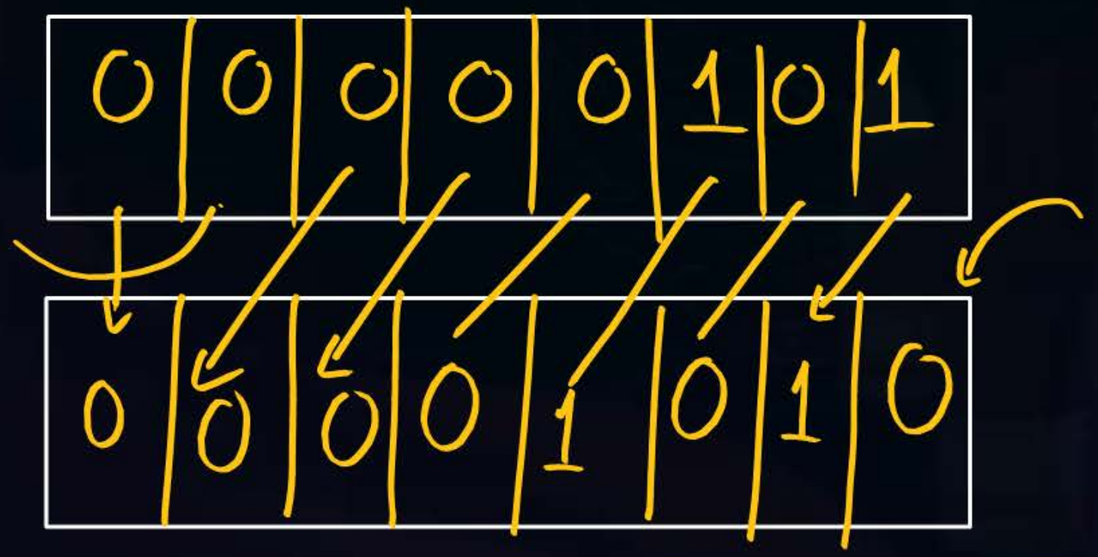
a ^ b

00000101

00000100

00000001

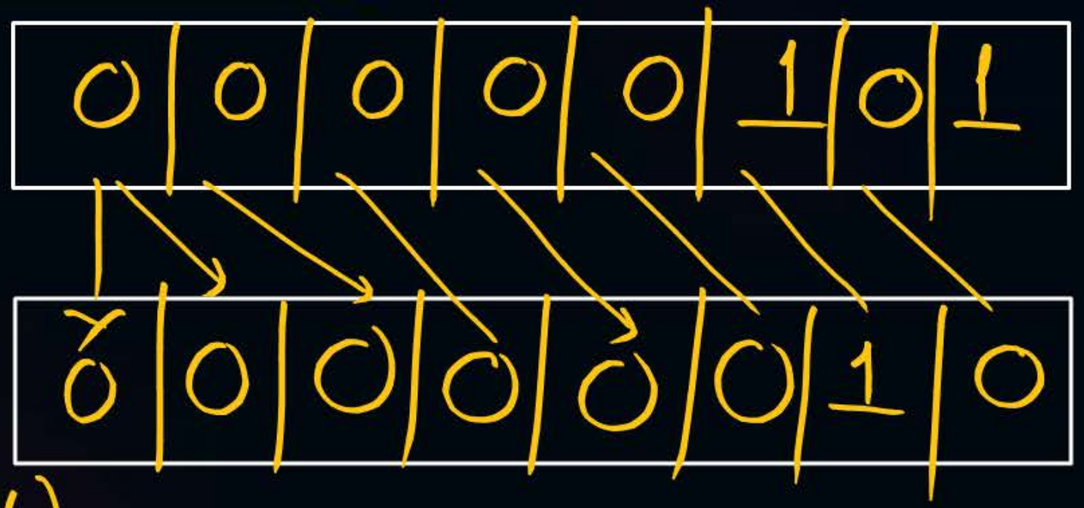
a << 1 ,



Slide

Bit-wise Operator

a >> 1




```
int a=4;
int b=5;
printf("%d", a&b);
printf("%d", a|b);
printf("%d", a^b);
```

Slide



2 mins Summary

topic Arithmetic operator

topic Relational

topic Logical

topic Bit wise

topic

Slide

THANK - YOU